

2023



# Third Space Learning

Learn how Third Space Learning in education reduced connectivity and audio issues while improving availability and scalability using Amazon Chime.

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## 50% reduction of in-session audio issues

sodales felis pellentesque et

## Achieved high availability

Pellentesque quis dui vel nunc cursus.

## Improved observability

Nulla nisl massa, ullamcorper id

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 **Increased scalability**  
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[Third Space Learning](#), which provides remote one-on-one mathematics tutoring to schools across the United Kingdom and the United States, needed a cost-effective audio layer for its solution that was reliable, always available, and had good connectivity. Although Third Space Learning's original home-built approach provided remote connectivity, the rise in demand following the COVID-19 pandemic meant the team needed to increase the reliability and scalability of its solution.

The company was cloud based at its inception and had been using Amazon Web Services (AWS) for portions of its infrastructure. To solve for scalability and connectivity for the audio portion of its solution, Third Space Learning chose [Amazon Chime](#), a communications service used to meet, chat, and place business calls inside and outside an organization, all using a single application. By migrating its audio to Amazon Chime, Third Space Learning has increased scalability, achieved high availability, and reduced in-session audio issues by 50 percent.

## Opportunity | Using Amazon Chime to Solve Connectivity and Scalability for Third Space Learning

Third Space Learning connects primary and secondary schools in the United Kingdom and the United States with tutors across Sri Lanka and India to provide mathematics tutoring for its students and to improve student outcomes. The company is democratizing tutoring by making it more financially accessible and works with over 3,800 schools and over 150,000 students, providing tens of thousands of sessions per week. By opening up one-on-one tutoring to reach a broader marketplace, the company promotes an equitable and inclusive solution to the need for high-quality individual mathematics instruction.

Founded in 2013, Third Space Learning has been using AWS since its beginnings. It started out with its monolithic application deployed on a single instance in [Amazon Elastic Compute Cloud](#) (Amazon EC2), which offers secure and resizable compute capacity for virtually any workload. Over the years, the company has used more than 25 different solutions for its audio layer. These solutions often met either the company's connectivity or its availability goals, but not both. Then, after many schools transitioned to remote learning in response to the COVID-19 pandemic, Third Space Learning's customer base tripled. This brought a new challenge of scalability to the forefront, because its solution at the time could not scale to meet demand.

In late 2020, the company decided to start using [Amazon Chime SDK](#), which makes it straightforward for builders to add near-real-time voice, video, and messaging powered by machine learning into their applications. "When Amazon Chime SDK was announced, we jumped on it as a solution for reliably scaling our usage while maintaining schools' ability to connect to the service," says Sam Stagg, chief technology officer at Third Space Learning at the time of the migration. This migration occurred alongside the company rolling out a new virtual classroom, a process that took 18 months. Third Space Learning completed the migration in 2021.

## Solution | Reducing In-Session Audio Issues by 50 Percent Using Amazon Chime

The spring term of 2023 was the first time that all of Third Space Learning's school clients were using the new virtual classroom on Amazon Chime. On the new infrastructure, the company saw the fewest reported audio issues in its history. "Using Amazon Chime SDK has been transformative for us," says Stagg. "It has been a positive experience virtually from start to finish." Overall, the company experienced a 50 percent reduction in reported audio issues during sessions throughout the spring term of 2023.

This benefit ties directly into improved connectivity and the simplicity with which Third Space Learning can now diagnose issues as they arise. Under the company's previous solution, it could not be sure why a connectivity lapse happened and would often have to refund customers for failed sessions. In addition, schools with more connectivity issues were less likely to renew with Third Space Learning at the end of the year.

Because the client-side data the company ingests from Amazon Chime is more in depth than what it gathered from its previous, self-hosted solution, Third Space Learning can more quickly identify the cause of connectivity issues. The company is using this data to help schools resolve connectivity issues and proactively engage with schools that have less connectivity. "Observability is a benefit for us because, previously, we couldn't always trust our system," says Stagg. "Using Amazon Chime, not only can we get to the bottom of an issue, but we can also fix it where before we couldn't."

Another benefit Third Space Learning has achieved is cost transparency. The company is now factoring in baseline costs over the year for running infrastructure and doing maintenance, and it is calculating lost opportunity costs. When it previously had to cancel sessions because its audio service went down, the company would credit customers for a future session, which also impacted the business's bottom line. Further, Third Space Learning no longer has to pay to spin up its audio solution for sessions that a school cancels. Instead, it has to pay only for the time during which it's actually using its solutions on AWS. "Our AWS use has grown considerably, but our spend is now much more proportionate to when services are actually used, which is a good thing to see," says Stagg.

## Outcome | Expanding Reach and Opportunities Using AWS

Now, in the second half of 2023, Third Space Learning is in the process of migrating its monolithic application to [Amazon Elastic Kubernetes Service](#) (Amazon EKS), a managed Kubernetes service used to run Kubernetes on AWS, to further improve the scalability of its solution. Scalability is important for the company as it has begun to expand its offerings to the United States and provide 24-hour tutoring services across multiple time zones. The company is working toward a future where its entire infrastructure is completely tracked in one place and it has the observability to identify and solve virtually any connectivity issue swiftly.

"The risk of what might happen to our audio service when we scale up is off the table," says Stagg. "We know that whatever we throw at Amazon Chime, even if we're successful beyond our wildest dreams in the United States, we won't be worrying about audio falling over."



# About Company

Third Space Learning works with schools in the United Kingdom and the United States to remotely connect students with mathematics-specialist tutors and provide one-on-one mathematics instruction. The company is also one of the larger providers of the UK's National Tutoring Programme.

## AWS Services Used

### Amazon Chime

Amazon Chime is a communications service that lets you meet, chat, and place business calls inside and outside your organization, all using a single application.

[Learn more »](#)

### Amazon Chime SDK

With the Amazon Chime SDK, builders can easily add real-time voice, video, and messaging powered by machine learning into their applications.

[Learn more »](#)

### Amazon Elastic Kubernetes Service (Amazon EKS)

Amazon Elastic Kubernetes Service (Amazon EKS) is a managed Kubernetes service to run Kubernetes in the AWS cloud and on-premises data centers.

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### Amazon Elastic Compute Cloud (Amazon EC2)

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute power in the cloud, with over 750 instances and choice of the latest processor, storage, networking, operating system, and pricing model to help you best match the needs of your workload.



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